

Nicolas Meseguer Iborra

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EDUCATION

- **Universidad de Murcia** Murcia, Spain
Doctor of Philosophy (Ph.D.) - Computer Architecture September 2022 - June 2027
Goal: Suggest approaches for addressing two transcendental concerns in the design of a heterogeneous processor's memory system; (1) novel organizations for the memory structures and (2) share the same memory space across general purpose cores and hardware accelerators.
- **Universidad de Murcia** Murcia, Spain
Master of Science (M.S.) - New Technologies in Computer Science; GPA: 9.24 September 2021 - June 2022
Courses: High Performance Computing and Architecture, Microarchitecture of the processing cores, Parallel programming standards, Compiler-optimised libraries, High performance system administration
- **Universidad Catolica San Antonio de Murcia** Murcia, Spain
Intern Student October 2019 - June 2020
Objectives: Introduction to Machine Learning (ML) and Deep Learning (DL) using novel-models: MLPs, RNNs and CNNs
- **Universidad Catolica San Antonio de Murcia** Murcia, Spain
Bachelor of Science (B.S.) - Computer Engineering; GPA: 9.31 October 2018 - June 2021
Courses: Computer Programming, Information Science, Networking, Operating Systems, Databases, Mathematics, Software Architecture, Algorithms, Data Structures
- **IES Jose Planes** Murcia, Spain
Higher National Certificate (HNC) - Web App Developer; GPA: 8.84 September 2016 - June 2018
Courses: Databases, High-level Computer Architecture, Web Language, Client and Server side frameworks, Usability and UX/UI

HONORS AND AWARDS

- Best Poster Award Winner (Jury Selection). Selected as one of 10 university-wide representatives for the University of Murcia at the XIII G9 Doctoral Symposium - May, 2026
- Competitive Pre-Doctoral Research Fellowship, at the AALP Lab at Purdue University under the supervision of Prof. Tim Rogers - Apr, 2026
- Third place in Hack4Good 24' with a comprehensive telemedicine platform with several stakeholders - Mar, 2024
- Programa Séneca 2022, research contract at the University of Murcia - Dec, 2022
- Second place in Mercedes-Benz Talent Highway with a predictive analytical market model - Nov, 2022
- University scholarship for the master's degree (M.S.) - Nov, 2022
- Telefónica Hack4Good 22' Winner, with an application against zero food waste - Oct, 2022
- Extraordinary Award for the Best Academic Record in Computer Engineering (B.S. Degree) - Jun, 2022
- Research Initiation Grant for promising young researchers (B.S. Degree) - Feb, 2021
- Erasmus+ Scholarship for a european internship (HNC Degree) - Mar, 2018

EXPERIENCE

- **Universidad de Murcia** Murcia, Spain
Fundación Séneca Researcher (Full-time) Mar 2023 - Present
 - **Objectives:** Suggest approaches for addressing two transcendental concerns in the design of a heterogeneous processor's memory system. 1) novel organizations for the memory structures employed by cores. 2) investigate alternatives to present virtual memory concepts in order to more efficiently share the same memory space.
- **Purdue University** West Lafayette, Indiana, USA
Visiting Scholar Researcher (Full-time) Aug 2026 - Dec 2026
 - **Objectives:** Proposing novel GPU microarchitectural improvements to optimize memory system efficiency and overall performance for state-of-the-art AI/ML workloads (e.g., LLMs, FlashAttention, NCCL) using Accel-Sim/GPGPU-sim.
- **Mowatave** Murcia, Spain
Chief Technology Officer Jan 2023 - Present
 - **Objectives:** Strategic technology development, technical leadership and innovation, ensuring security and compliance, fostering communication and collaboration, and promoting staff training and development.
- **Universidad de Murcia** Murcia, Spain
Postgraduate extracurricular research internship (Part-time) Nov 2022 - Dec 2022
 - **Objectives:** Recently a GPU architecture simulator (MGPU-Sim) has been publicly released and the research group wants to add it to its catalog of simulation tools for research. The student will have to install and configure the simulator in the group's computing cluster, as well as prepare the necessary software infrastructure.

- **Universidad Catolica San Antonio de Murcia** Murcia, Spain
Research Initiation Grant (Part-time) Feb 2021 - Jul 2021
 - **Objectives:** In charge of developing a hybrid accelerator (CPU+GPU) for a recommender neural network model (DLRM, WATERoT) using specialized accelerator architectures for inference (SIGMA, MAERI). Aiming at maximizing the model efficiency, waiting times and studying memory accesses (W/R) and total cycles consumed.
- **Plinga GmbH** Berlin, Germany
Full-stack Web Developer Erasmus+ Internship (Full-time) Apr 2018 - Sep 2018
 - **Experience acquired:** First-hand experience on front-end and back-end of various PHP and Ruby projects, including Amazon AWS services (EC2 servers, Databases and CDNs)

PUBLICATIONS

- Nicolas Meseguer, Daoxuan Xu, Yifan Sun, Michael Pellauer, José L. Abellán, and Manuel E. Acacio. “**QuCo: Efficient and Flexible Hardware-Driven Automatic Configuration of Tile Transfers in GPUs**”. *32nd IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. Sydney, Australia, February 2026.
- Nicolas Meseguer, Yifan Sun, Michael Pellauer, José L. Abellán, and Manuel E. Acacio. “**Configuración Automática del TMA para GPUs de Alto Rendimiento**”. *Work in XXXV Jornadas de Paralelismo (JP)*. Seville, Spain, June 2025.
- Nicolas Meseguer, Yifan Sun, Michael Pellauer, José L. Abellán, and Manuel E. Acacio. “**ACTA: Automatic Configuration of the Tensor Memory Accelerator for High-End GPUs**”. *17th Workshop on General Purpose Processing Using GPU (GPGPU)*. Las Vegas, USA, March 2025.
- Nicolas Meseguer, José L. Abellán, and Manuel E. Acacio. “**CPU-PCGCN: Efficient Processing of Convolutional Graph Networks on CPU Architectures**”. *Poster in 19th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems (ACACES)*. Fiuggi, Italy, July 2023.
- Nicolas Meseguer, José L. Abellán, and Manuel E. Acacio. “**CPU-PCGCN: Procesamiento Eficiente de Redes Convolucionales de Grafos en Arquitecturas CPU**”. *Work in XXXII Jornadas de Paralelismo (JP)*. Alicante, Spain, September 2022.
- Nicolas Meseguer, José L. Abellán, and Manuel E. Acacio. “**Evaluacion de un Sistema de Recomendacion en un Acelerador Híbrido**”. *Work in XXXI Jornadas de Paralelismo (JP)*. Malaga, Spain, September 2021.

ACADEMIC EXPERIENCE

- **Multi-core and Purpose-built Architectures** University of Murcia
Teaching Assistant – 2.50 ECTS February 2026 - June 2026
- **Introduction to Operating Systems** University of Murcia
Teaching Assistant – 2.50 ECTS September 2025 - January 2026
- **Multi-core and Purpose-built Architectures** University of Murcia
Teaching Assistant – 2.50 ECTS February 2025 - June 2025
- **Introduction to Operating Systems** University of Murcia
Teaching Assistant – 2.50 ECTS September 2024 - January 2025
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- **Multi-core and Purpose-built Architectures** University of Murcia
Teaching Assistant – 2.50 ECTS February 2023 - June 2023

STUDENTS

- **Nuria Martinez Albuquerque** 9.9 Honorable Mention
Implementation and Evaluation of Data Prefetching Schemes on a GPU Architecture September 2025 - June 2026
- **José Ángel Nicolás Ortega** 9.9 Honorable Mention
Register Bank Modeling on GPUs with MGPU-Sim September 2022 - June 2023